Question Paper: Machine Learning

Machine Learning Question Paper

Class 9-12

Medium Difficulty

Multiple Choice Questions (30 marks)

1. What is the primary goal of machine learning?

A) To develop algorithms for solving mathematical problems

B) To enable machines to learn from data and make predictions

C) To create expert systems that mimic human decision-making

D) To optimize computer hardware for faster processing

Answer: **B) To enable machines to learn from data and make predictions**

2. Which of the following is a type of supervised learning?

A) Clustering

B) Regression

C) Dimensionality reduction

D) Anomaly detection

Answer: **B) Regression**

3. What is the term for the process of adjusting model parameters to minimize the difference between predicted and actual outputs?

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A) Backpropagation

B) Gradient descent

C) Regularization

D) Cross-validation

Answer: **B) Gradient descent**

4. Which algorithm is commonly used for image classification tasks?

A) Decision Trees

B) Support Vector Machines

C) Convolutional Neural Networks

D) K-Means Clustering

Answer: **C) Convolutional Neural Networks**

5. What is the purpose of feature scaling in machine learning?

A) To reduce the dimensionality of the data

B) To improve the interpretability of the model

C) To prevent overfitting

D) To ensure that all features have the same range

Answer: **D) To ensure that all features have the same range**

6. Which of the following is a type of unsupervised learning?

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A) Linear Regression

B) Logistic Regression

C) K-Means Clustering

D) Decision Trees

Answer: **C) K-Means Clustering**

7. What is the term for the evaluation of a model on unseen data?

A) Training

B) Testing

C) Validation

D) Cross-validation

Answer: **B) Testing**

8. Which of the following is a deep learning technique?

A) Linear Regression

B) Decision Trees

C) Support Vector Machines

D) Recurrent Neural Networks

Answer: **D) Recurrent Neural Networks**

9. What is the purpose of regularization in machine learning?

A) To prevent overfitting

B) To improve the interpretability of the model

C) To reduce the dimensionality of the data

D) To increase the computational efficiency of the model

Answer: **A) To prevent overfitting**

10. Which of the following is a type of ensemble learning method?

A) Bagging

B) Boosting

C) Stacking

D) All of the above

Answer: **D) All of the above**

Short Questions (15 marks)

1. What is the difference between supervised and unsupervised learning? (3 marks)

2. Explain the concept of overfitting in machine learning. (3 marks)

3. Describe the process of data preprocessing in machine learning. (4 marks)

4. What is the purpose of feature selection in machine learning? (3 marks)

5. Explain the concept of bias-variance tradeoff in machine learning. (4 marks)

Long/Descriptive Questions (30 marks)

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1. Describe the different types of machine learning algorithms, including supervised, unsupervised, and reinforcement learning. Explain the strengths and weaknesses of each type. (10 marks)
2. Explain the concept of neural networks and how they are used in machine learning. Describe the different types of neural networks, including feedforward, recurrent, and convolutional neural networks. (10 marks)
3. Discuss the importance of model evaluation in machine learning. Explain the different metrics used to evaluate the performance of a model, including accuracy, precision, recall, and F1 score. Describe the process of cross-validation and its benefits. (10 marks)

Answer Key

Multiple Choice Questions

1. B) To enable machines to learn from data and make predictions
2. B) Regression
3. B) Gradient descent
4. C) Convolutional Neural Networks
5. D) To ensure that all features have the same range
6. C) K-Means Clustering
7. B) Testing
8. D) Recurrent Neural Networks
9. A) To prevent overfitting
10. D) All of the above

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Short Questions

1. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data.
2. Overfitting occurs when a model is too complex and performs well on the training data but poorly on unseen data.
3. Data preprocessing involves cleaning, transforming, and feature scaling the data to prepare it for modeling.
4. Feature selection involves selecting the most relevant features to include in the model to improve its performance.
5. The bias-variance tradeoff refers to the tradeoff between the accuracy of the model and its ability to generalize to new data.

Long/Descriptive Questions

1. Machine learning algorithms can be broadly classified into supervised, unsupervised, and reinforcement learning. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data. Reinforcement learning involves training a model to take actions to maximize a reward. Each type of algorithm has its strengths and weaknesses, and the choice of algorithm depends on the specific problem and data.
2. Neural networks are a type of machine learning algorithm inspired by the structure and function of the human brain. They consist of layers of interconnected nodes or neurons that process and transmit information. Feedforward neural networks are the simplest type of neural network, while

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recurrent neural networks are more complex and can handle sequential data. Convolutional neural networks are a type of neural network that is particularly well-suited for image classification tasks.

3. Model evaluation is an essential step in machine learning, as it allows us to assess the performance of the model and identify areas for improvement. Metrics such as accuracy, precision, recall, and F1 score are commonly used to evaluate the performance of a model. Cross-validation is a technique that involves splitting the data into training and testing sets and evaluating the model on the testing set. This helps to prevent overfitting and ensures that the model generalizes well to new data.